

Table 1: Current P0 generation comparison. Lower Frechet-style metrics are better.

Dataset	C low	D low	DDPM low	EDM low	low winner	C Inc	D Inc	DDPM Inc	EDM Inc	Inc winner	C rank
CIFAR-10	7.27	6.43	7.86	2.87	EDM	313.89	320.14	498.65	351.17	CoFiTok	1
Tiny-IN	8.22	8.27	9.07	3.10	EDM	373.24	385.51	381.24	268.72	EDM	2
ImageNet-64 HF	8.53	7.67	8.88	3.23	EDM	374.35	388.15	378.33	258.40	EDM	2
ImageNet-64 strict	8.36	7.58	8.80	2.75	EDM	374.68	377.12	355.11	248.31	EDM	3
FFHQ-64	8.74	8.09	9.44	2.51	EDM	425.71	424.11	369.01	325.49	EDM	4
AFHQv2-64	7.08	6.58	8.29	1.96	EDM	331.06	345.94	335.18	230.40	EDM	2
ImageNet-256 10%	10.44	10.51	10.40	10.39	EDM	414.19	337.87	419.60	368.68	Dense epsilon	3
ImageNet-256	10.54	10.28	10.40	10.40	Dense epsilon	417.26	354.43	414.30	368.43	Dense epsilon	4

C = CoFiTok; D = same-backbone dense epsilon. These are current smoke/short-run Frechet-style metrics, not formal SOTA FID claims.

Table 2: Diagnostics for the supported CoFiTok claim.

Dataset	C PSNR	D PSNR	mask PSNR	C path	D path	C effK	D effK	C zero	deep-S zero
CIFAR-10	14.118	14.390	12.938	0.0960	1.3022	6.85	5.16	0.0000	0.0619
Tiny-IN	14.250	14.375	12.955	0.0763	0.8459	6.84	5.38	0.0000	0.0505
ImageNet-64 HF	15.062	15.131	13.593	0.0641	0.8837	6.77	5.40	0.0000	0.0450
ImageNet-64 strict	14.521	14.651	13.250	0.0681	1.0403	6.82	5.38	0.0000	0.0535
FFHQ-64	15.024	15.144	13.285	0.0682	1.0807	6.82	5.42	0.0000	0.0518
AFHQv2-64	15.167	15.196	13.384	0.0643	1.0118	6.82	5.46	0.0000	0.0561
ImageNet-256 10%	13.627	13.867	13.807	0.1347	1.5609	3.35	3.66	0.0000	0.0179
ImageNet-256	13.809	13.877	13.761	0.1255	1.4917	3.37	3.66	0.0000	0.0185

Lower path AUC indicates better prefix controllability. Zero-token ratio diagnoses whether the synthesis operator leaks structure at zero input.

Table 3: P1 eval-only tokenizer reconstruction baselines.

Dataset	src	eval	FlexTok PSNR	FlexTok low	TiTok PSNR	TiTok low
AFHQv2-64	64	256	24.04	0.036	17.29	0.217
CIFAR-10	32	256	26.77	0.037	19.35	0.168
ImageNet-64 strict	64	256	22.57	0.042	16.73	0.196
FFHQ-64	64	256	24.06	0.042	17.39	0.231
ImageNet-64 HF	64	256	23.60	0.039	18.04	0.182
ImageNet-256	256	256	19.12	0.044	15.54	0.196
ImageNet-256 10%	256	256	19.12	0.044	15.54	0.196
Tiny-IN	64	256	21.81	0.044	17.30	0.160

Official pretrained reconstruction at 256x256 on 256 images per dataset. These rows are not retrained generation baselines.

Table 4: Official related-method eval-only rows.

Method	Dataset	Status	Samples	FID	sFID	IS	Prec.	Rec.
D-AR	ImageNet-256	completed_eval_only_50k	50000	2.628	6.720	285.89	0.803	0.588
MAR	ImageNet-256	sampling_or_eval_pending	3072	—	—	—	—	—
ReTok	ImageNet-256	completed_eval_only_50k	50000	2.219	5.896	245.94	0.816	0.599

These rows use official pretrained protocols where available and are secondary comparisons; they are not part of the all-dataset same-budget P0 training matrix.

Table 5: Claim support summary.

Claim axis	Current evidence
Generation quality	CoFiTok is best on 0/8 lowres rows and 1/8 Inception rows; average ranks are 3.00 and 2.62. Do not claim generation SOTA.
Prefix control	CoFiTok path AUC is lower than dense on 8/8 datasets.
Repeated 20k evidence	CoFiTok lowers path AUC in 4/4 paired dataset-seed runs (mean relative reduction 96.31%), while endpoint MSE changes by +2.89% on average.
Restricted synthesis	CoFiTok zero-token ratio is zero in current rows; deep- S_k has nonzero zero-token ratio on 8/8 datasets.
Related methods	completed official 50K: D-AR, ReTok; running/eval pending: MAR. All rows remain secondary and outside P0 fair training.
Main paper stance	The supported claim is ordered restricted dense-noise factorization with prefix-controllable denoising, not broad unconditional generation superiority.